Exercise Set 2

Prof. Kenny Paterson
K. T. Truong, S.-P. Merz, Y. Song

In this exercise sheet, we will introduce the Schnorr signature scheme, which has been used in Bitcoin since the Taproot update in 2021 due to its nice algebraic properties. We will then show that one can perform a key recovery attack on this signature scheme if information is given about the nonces used in the signing process. We will first show some simple attacks based on known and repeated nonces. We will then cast the problem of finding the private key given the most significant bits of a few nonces as an instance of the Hidden Number Problem (HNP), which we will reduce to an instance of the Closest Vector Problem (CVP), seen in the lectures. This is one of the most important applications of lattices in cryptanalysis, and is a good example of how to use the tools from the lectures to attack a real-world cryptographic scheme.

Focus during the exercise session: During the exercise session, we will not cover Problem 1, except for the description of the Schnorr signature scheme in Part 1.c. We suggest you work through Problem 1 on your own, so you understand the logic behind the scheme, without having to take the formulae for granted.

For Problem 2, we will focus on the intuition needed to understand the attack, specifically Parts 2.a, 2.c, 2.d, 2.f and 2.g. We will mostly skip the computations, and we strongly suggest you work through them on your own.

Problem 1. Throughout this entire exercise sheet, let $\mathbb{G}$ be a cyclic group of prime order q with generator P, in which computing discrete logarithms is assumed to be hard. We will be working over the group of points of an elliptic curve, therefore we will be working with the additive notation for the group operation: we will write $[x]P$ for the group element obtained by adding P to itself x times. The group description $(\mathbb{G}, q, P)$ is public. A secret key is an element $x \in \mathbb{Z}_q$, and the corresponding public key is $X = [x]P \in \mathbb{G}$.

We begin with an interactive protocol – *Schnorr's Identification Scheme* – where a prover $\mathcal{P}$ holding the secret key x wants to convince a verifier $\mathcal{V}$, who only holds the public key X, that it knows the discrete logarithm of X to the base P, without revealing anything else about x. We will later apply a transformation to turn this protocol into a signature scheme. The protocol consists of the following three-move interaction:

Prover $\mathcal{P}(P, X, x)$		**Verifier** $\mathcal{V}(P, X)$
$k \leftarrow\!\!\$\ \mathbb{Z}_q$		
$R \leftarrow [k]P$		
	$\xrightarrow{R \text{ (commitment)}}$	
		$t \leftarrow\!\!\$\ \mathbb{Z}_q$
	$\xleftarrow{t \text{ (challenge)}}$	
$s \leftarrow k - xt \pmod q$		
	$\xrightarrow{s \text{ (response)}}$	
		accept iff $[s]P + [t]X = R$

(a) Show that this protocol is complete, i.e. that an honest prover always convinces the verifier.

(b) Show that if the prover can predict the verifier's challenge t before sending the commitment R, then it can compute a valid response s without knowing the secret key x.

(c) One can show that this protocol leaks nothing about x to a verifier that follows the protocol honestly and that a cheating prover that does not know x can only convince the verifier with probability $1/q$. We will not prove these properties here, but they are crucial for the security of the signature scheme we will derive from this protocol. What we need to convert this protocol into a signature scheme is to make it *non-interactive*. To this end, we use the *Fiat-Shamir transform*. Assume we want to sign a message m, then we replace the verifier's random challenge t with $H(R \,\|\, m)$. This allows the signer to honestly generate a challenge without interacting with the verifier. The resulting signature scheme is given below.

Schnorr Signature Scheme. Let $\mathbb{G}$ be a cyclic group of prime order q with generator P, as above, and let $H : \{0,1\}^* \to \mathbb{Z}_q$ be a hash function. The Schnorr signature scheme is defined as follows:

KeyGen	Choose a random $x \in \mathbb{Z}_q$ and compute $X = [x]P$. The public key is $pk = X$ and the secret key is $sk = x$.
Sign	To sign a message $m \in \{0,1\}^*$, choose a random nonce $k \in \mathbb{Z}_q$ and compute $R = [k]P$. Compute $t = H(R \,\|\, m)$ and $s = k - xt \pmod q$. The signature is $\sigma = (s, t)$.
Verify	To verify a signature $\sigma = (s, t)$ on a message m, compute $R' = [s]P + [t]X$ and $t' = H(R' \,\|\, m)$. The signature is valid if and only if $t' = t$.

Informally argue why this makes the scheme non-interactive, while preserving the security properties of the original protocol.

(d) Show that, if an adversary knows the value of the nonce k used to sign a message m, then it can compute the secret key x (except with negligible probability).

(e) Show that, if the signer uses the same nonce k to sign two different messages m_1 and m_2, then an adversary can compute the secret key x (assuming no hash collisions happen).

Problem 2. We now show that if the most significant bits of the nonce k are known to an adversary, then it can compute the secret key x after seeing enough signatures. For simplicity, we will analyze the case where we assume that the L most significant bits of k are zero.[1] Let $N = \lceil \log_2 q \rceil$ be the bit-length of q. Then, we can write

$$k = 2^{N-L-1} + \varepsilon$$

where ε is a small error term, i.e. $|\varepsilon| \leq 2^{N-L-1}$. Next consider the signature equation:

$$s = k - xt \pmod q$$

which can be rewritten as

$$xt = k - s = 2^{N-L-1} - s + \varepsilon \pmod q.$$

Set $u = 2^{N-L-1} - s$ to obtain:

$$xt = u + \varepsilon \pmod q$$

[1] One can easily extend to the case where the MSBs are known but not necessarily zero.

with $|\varepsilon|$ small and t and u known. In other words, we have that u is a "good" approximation of xt modulo q. The problem of finding x given many such approximations is known as the *Hidden Number Problem* (HNP). That is, given n pairs (t_i, u_i) such that $xt_i = u_i + \varepsilon_i \pmod{q}$ with $|\varepsilon_i|$ small, find x.

Our strategy will be to reduce the HNP to an instance of the Closest Vector Problem (CVP) in a lattice, where the target vector is composed of publicly known values and the closest vector contains the secret key x in its last coordinate.

(a) Consider the following basis matrix B for a lattice $\mathcal{L}_B$:

$$B = \begin{bmatrix} q & 0 & 0 & \cdots & 0 & 0 \\ 0 & q & 0 & \cdots & 0 & 0 \\ 0 & 0 & q & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & q & 0 \\ t_1 & t_2 & t_3 & \cdots & t_n & 1/2^{L+1} \end{bmatrix}$$

and let $\underline{u} = (u_1, u_2, \ldots, u_n, 0)$. Prove that there exists a vector $\underline{v}$ in $\mathcal{L}_B$ such that:

$$\|\underline{u} - \underline{v}\| < \sqrt{n+1} \cdot 2^{N-L-1}.$$

Hint: Let $\underline{v} = (-\ell_1, -\ell_2, \ldots, -\ell_n, x) \cdot B$, where each ℓ_i is the integer such that $xt_i = u_i + \varepsilon_i + \ell_i q$ (over $\mathbb{Z}$).

(b) Recall the Gaussian heuristic from the lecture: this states that, for a "random" lattice $\mathcal{L}$ in n' dimensions, we have:

$$\lambda_1 \approx \sqrt{\frac{n'}{2\pi e}} \cdot \det(\mathcal{L})^{1/n'}.$$

Compute the determinant of B and the expected value of λ_1 for the corresponding lattice (under the Gaussian heuristic).

(c) Compare the distance $\|\underline{u} - \underline{v}\|$ from Part 2.a to the expected value of λ_1 from Part 2.b, showing that $\underline{u}$ and $\underline{v}$ are abnormally close. Then, looking at the last coordinate of $\underline{v}$, explain how to recover the secret key x assuming we have an algorithm that solves the CVP for the lattice $\mathcal{L}_B$.

Hint: Check that for large enough n, $\|\underline{u} - \underline{v}\| < \lambda_1(\mathcal{L}_B)/2$. It might be helpful to use the approximations $x < q \approx 2^N$, $\sqrt{2\pi e} \approx 2^{2.05}$ and work by taking logarithms.

(d) Recall that, in the Kannan embedding approach to solving CVP, we augment the lattice B to obtain:

$$B' = \begin{bmatrix} B & 0 \\ \underline{u} & M \end{bmatrix}.$$

Here $\underline{u}$ is the target input vector for CVP. Let $\underline{v}$ be the solution to the CVP problem. A shortest vector in the lattice $\mathcal{L}' = \mathcal{L}_{B'}$ is expected to be of the form $(\underline{f}, M)$ where $\underline{f} = \underline{u} - \underline{v}$. Compute an upper bound on the norm of $(\underline{f}, M) \in \mathcal{L}'$ (as a function of M).

(e) Compute the determinant of B' and the expected length of the shortest vector λ_1' for the corresponding lattice (under the Gaussian heuristic).

(f) Compare the estimate of λ_1' from Part 2.e to the bound on $\|(\underline{f}, M)\|$ you obtained in Part 2.d. What does this suggest about how to set M in order to ensure the success of the Kannan embedding approach to solving CVP in this case?

(g) Consider $\underline{z} = (-t_1, -t_2, -t_3, \ldots, -t_n, q, 0) \cdot B' \in \mathcal{L}'$. Compute the norm of $\underline{z}$ and compare it to the bound on the norm of $(\underline{f}, M)$ you obtained in Part 2.d, showing that it is typically about $\sqrt{n+1}$ times smaller (irrespective of the choice of M).

This problem shows that $\mathcal{L}'$ contains unusually short vectors, so solving SVP may not directly yield a solution to the original CVP for the particular lattices we have constructed here.

On the other hand, perhaps the required solution will appear *elsewhere* in the basis obtained from performing lattice reduction on B'...